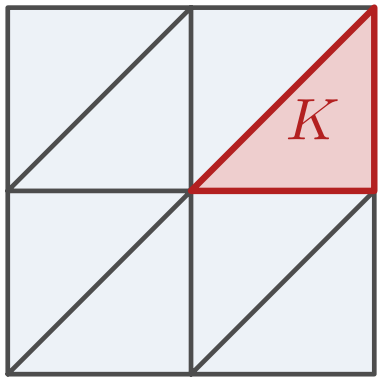
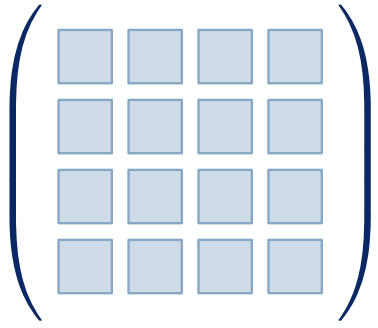


mesh $\mathcal{T}_h$

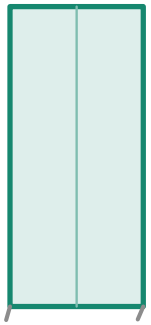
$W_K \in \mathbb{R}^{n\times n}$



$(W_K)_{ij} = (\mathcal{A}\varphi_j, \mathcal{A}\varphi_i)_K$

SVD

T_K



$n \times m, \;\; m \ll n,$
columns spanning $\ker_\varepsilon(W_K)$

$$T = \operatorname{diag}\big(T_K\big)_{K\in\mathcal{T}_h} \in \mathbb{R}^{N\times M}$$

$$\Rightarrow \quad T^\top A T\, \boldsymbol{u}_T = T^\top \boldsymbol{\ell}$$

